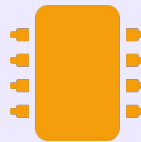


Problem Statement & Objectives



The Gap

Bridging visual understanding with natural language generation is one of AI's core challenges, from accessibility tools to content indexing.



Compute Barrier

Models like Flamingo & BLIP train on billions of image-text pairs and require thousands of GPU hours — inaccessible for most.



Our Objective

Design, implement & optimize an image-to-text pipeline integrating CLIP + GPT-2 under strict resource and data constraints.

Related Work



Traditional Captioning

01

- › CNN-LSTM encoder-decoder (Vinyals et al.)
- › Spatial attention: focus on image regions (Xu et al.)
- › Suffered from sequential RNN bottleneck



Vision-Language Pretraining

02

- › ViT: image patches as tokens (Dosovitskiy et al.)
- › CLIP & ALIGN: contrastive learning on image-text pairs
- › CLIP = powerful encoder, but cannot generate text



Large Generative VLMs

03

- › Flamingo, BLIP: vision encoder + LLM via cross-attention
- › GIT: unified vision encoder + text decoder (inspires us)
- › Require billions of pairs & thousands of GPU-hours



Parameter-Efficient FT (PEFT)

04

- › Prefix-tuning: trainable vectors in attention (Li et al.)
- › Adapter modules: bottleneck layers (Houlsby et al.)
- › LoRA: low-rank matrices, ~1% params (Hu et al.)

Dataset & Preprocessing

2,500

Training Images

COCO Captions subset

12,500

Training Pairs

5 captions per image

224×224

Image Resolution

Resized & normalized

Processing Pipeline

Images

Resized 224×224px
Normalized for CLIP ViT

Captions

GPT-2 tokenizer
Start/end tokens added

Alignment

5 ground-truth captions
per image for richness

Model Architecture

01



Vision Encoder

CLIP ViT (Frozen)

Pre-trained CLIP ViT extracts 768-dim latent representations. Kept frozen to preserve generalized visual knowledge from millions of images.

02



The Bridge

MLP Projection Layer

Linear → GELU → Linear maps CLIP's 768D embedding into GPT-2's embedding space, acting as the cross-modal translator.

03



Language Decoder

GPT-2 + LoRA

GPT-2 receives projected image as prefix token. Auto-regressively generates caption token-by-token conditioned on visual prefix.

IMAGE → CLIP ViT → MLP Bridge → GPT-2 (LoRA) → CAPTION

Fine-Tuning Strategies

MLP-Only (Prefix Tuning)

~0% GPT-2 updated

✓ Stable convergence
No catastrophic forgetting

✗ Limited capacity
Descriptive ceiling

Both CLIP and GPT-2 frozen. Only the MLP bridge trains. Val loss: 3.06 → 2.74 (plateau).

Baseline

Full Fine-Tuning

~117M params trained

✓ Maximum flexibility

✗ Severe overfitting (epoch 4+)
Context collapse & hallucinations

All GPT-2 params unfrozen. Large train/val gap. Memorizes training patterns.

Risky

Low-Rank Adaptation (LoRA)

~1% params (rank $r=8$)

✓ Best metrics across all
Implicit regularizer
Preserves GPT-2 syntax

✗ Slightly more setup

Injects trainable A,B matrices into Q,V attention. $\alpha=16$. Stable minimal train/val gap.

★ Optimal

Decoding Strategies



Greedy Search

Picks highest-probability token at each step.

✓ Fast, deterministic

✗ Myopic — misses globally better sequences

e.g. "man standing on beach surfboard"



★ Beam Search (k=5)

Explores 5 parallel sequence paths simultaneously.

✓ Best structural accuracy & visual grounding

✗ Less creative; higher compute

e.g. "man standing on top of a surfboard in the middle of an ocean"



Nucleus (Top-p) Sampling

Samples from tokens whose cumulative prob $\geq p=0.9$.

✓ Creative, diverse outputs

✗ Drifts from visual ground truth — lower CIDEr/BLEU

e.g. "Friends chilling outdoors together near the water"

Temperature scaling also controls diversity — higher $T \rightarrow$ more varied, lower $T \rightarrow$ more conservative output.

Training Setup & Hyperparameters

Hardware & Optimizer

GPU	NVIDIA RTX 3090 (single)
Optimizer	AdamW
Learning Rate	2e-5 (LoRA/Full FT) 5e-5 (Prefix)
Batch Size	16
Epochs	10
Scheduler	Linear warmup (500 steps) + Cosine decay

LoRA Configuration

Rank (r)	8
Alpha (α)	16
Target Layers	Query & Value projections
Trainable %	~1% of total parameters

Evaluation Metrics

BLEU-1/4	METEOR	CIDEr	ROUGE-L
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BLEU: n-gram overlap • METEOR: semantic (stems + synonyms) • CIDEr: TF-IDF captioning metric • ROUGE-L: longest common subsequence

Full Quantitative Results

Model Config	Decoding	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
Prefix Tuning	Greedy	35.86	5.10	35.84	28.15	1.94
Prefix Tuning	Beam (k=5)	37.40	6.55	39.05	29.77	3.28
Prefix Tuning	Nucleus (p=0.9)	32.96	3.32	32.44	25.78	0.93
Full FT	Greedy	37.60	6.26	36.87	27.21	2.72
Full FT	Beam (k=5)	38.31	7.60	39.24	27.93	4.00
Full FT	Nucleus (p=0.9)	34.11	3.95	32.45	25.08	1.17
LoRA	Greedy	39.16	7.46	39.73	29.64	4.07
LoRA	Beam (k=5)	40.06	8.89	41.76	30.93	5.77
LoRA	Nucleus (p=0.9)	35.77	4.59	35.63	26.89	1.41

★ LoRA + Beam Search achieves best performance across all metrics on the 2,500-image constrained dataset.

Training Dynamics & Loss Curves

MLP-Only (Prefix Tuning)



Full Fine-Tuning (~117M params)



LoRA (~1% params) — Stable, no overfitting



MLP-Only:

Val loss plateaus at 2.74. Stable but capacity-limited — MLP bridge alone insufficient for complex alignment.

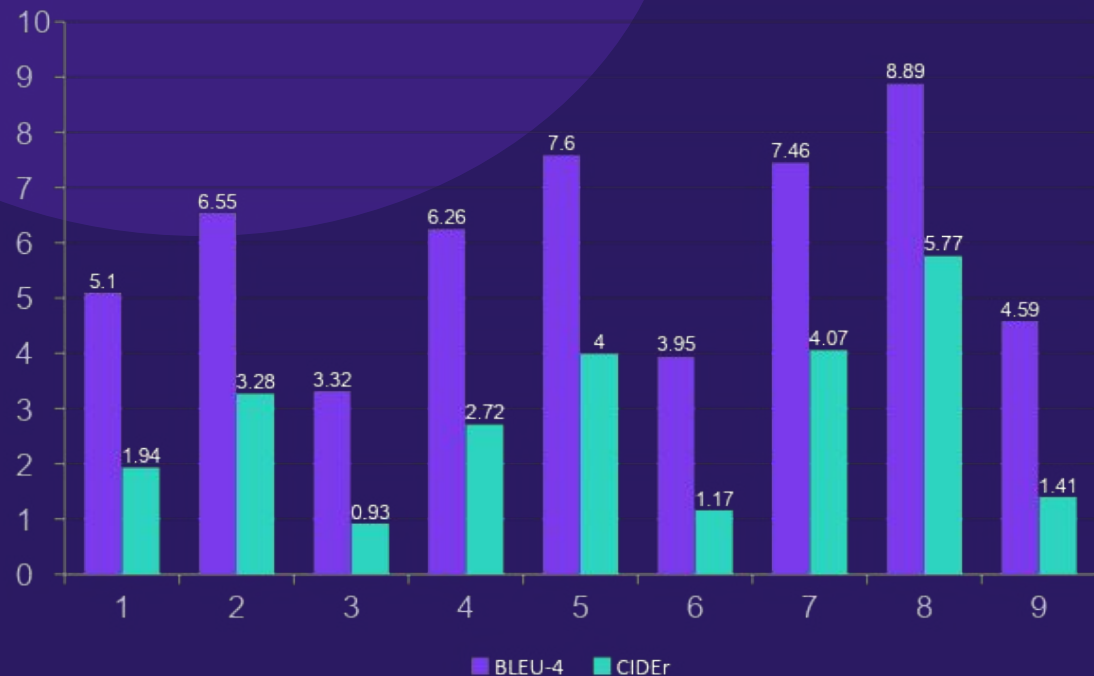
Full FT:

Large train/val gap after epoch 4. Classic overfitting — model memorises training set (context collapse).

LoRA:

Minimal stable train/val gap throughout. Convergence to ~2.60 without overfitting — best of both worlds.

Performance Comparison



8.89

BLEU-4 • LoRA + Beam

5.77

CIDEr • LoRA + Beam

41.76

METEOR • LoRA + Beam

30.93

ROUGE-L • LoRA + Beam

* Constrained 2,500-image training set. State-of-the-art on full COCO: BLEU-4 $\approx$ 30–40

Qualitative Evaluation — Generated Captions

Sample 1: Desert Scene

Ground Truth: "2 people walking on a desert laden with sand."

Prefix Tuning — Beam

"a man is standing on the ground with his hands up... it seems"

LoRA — Beam ★

"man standing on top of a surfboard in the middle of an ocean. He is holding onto his skis as he walks by them with other people"

Full FT — Beam

"man riding on top of a surfboard in the ocean. He is holding onto it while standing next to other people and trees nearby."

⚠ Both models misidentify desert dunes as ocean. Full FT hallucinates "trees" — absent from both contexts.

Sample 2: Children Painting

Ground Truth: "Children painting on a desk."

Prefix Tuning — Beam

"a group of children playing with their toys. The kids are sitting on the playground and they have some food in front of them..."

LoRA — Beam ★

"children playing with a skateboard at the park. The kids are dressed in colorful clothing and pose for pictures while they play..."

Full FT — Beam

"children playing in a park with umbrellas. One child is holding an umbrella and the other holds it up for everyone to see."

⚠ Full FT hallucinates 'umbrellas' & 'birthday cakes'. LoRA correctly identifies children + activity + appearance.

Analysis: Key Findings & "Context Collapse"



Context Collapse (Full Fine-Tuning)

When all 117M GPT-2 params are unfrozen, the model memorises training patterns. It hallucinates high-frequency concepts — "umbrellas", "trees", "cakes" — regardless of actual image content. Identified as a core failure mode of full FT on limited data.



LoRA as Implicit Regularizer

Restricting trainable params to ~1% forces generalised cross-modal mapping. Preserves GPT-2's pre-trained syntactic fluency. No catastrophic forgetting. Stable train/val loss gap confirms implicit regularisation.

LoRA Mathematics

$$h = W_0 x + \Delta W x = W_0 x + B A x \quad \text{where } W_0 \in \mathbb{R}^{(d \times k)}, \quad B \in \mathbb{R}^{(d \times r)}, \quad A \in \mathbb{R}^{(r \times k)}, \quad \text{rank } r \ll \min(d, k)$$

$r = 8, \alpha = 16$, applied to Query & Value projection layers of all GPT-2 attention blocks

Fidelity vs. Diversity

Beam Search: Best semantic fidelity — anchors language to visible subjects. Optimal for analytical tasks.

Nucleus Sampling: Creative diversity but drifts from visual ground truth → lower CIDEr/BLEU scores.

Greedy: Fast but myopic — misses globally optimal sequences.

Ethical Considerations, Limitations & Future Work

Ethical Concerns

Dataset Bias

COCO absorbs Western cultural contexts & default gender assumptions for ambiguous subjects.

Hallucination Risk

Invented objects in medical imaging or accessibility tools pose direct safety risks.

Privacy

Web-scraped data may include individuals without consent; deployed systems risk surveillance.

Limitations

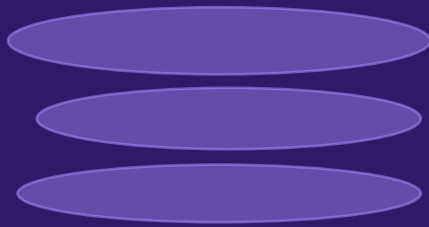
- › 2,500-image subset limits vocabulary diversity
- › English-only output (no multilingual support)
- › Static images only — no video / temporal processing
- › Single-sentence captions only

Future Directions

- › Scaling: expand to 10k–100k images to find Full FT crossover point
- › Replace GPT-2 with LLaMA / Mistral; CLIP with SigLIP or DINOv2
- › GradCAM attention visualisation for interpretability
- › Multi-caption & multi-language generation



Conclusion



Successfully bridged CLIP + GPT-2 for end-to-end image-to-text generation via prefix conditioning.



LoRA + Beam Search (k=5) is the optimal configuration: BLEU-4 8.89, CIDEr 5.77, METEOR 41.76, ROUGE-L 30.93.



Parameter-efficient fine-tuning is essential on constrained datasets — trains only ~1% of parameters.



Identified 'context collapse' as a core failure mode of full fine-tuning on limited data.

Thoughtful architecture + parameter efficiency → Robust generative AI, even under resource constraints.